



Human- and AI-based authorship: Principles and ethics

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Key points

- The International Committee of Medical Journal Editors (ICMJE) recommendations for authorship are the dominant guidelines that guide who, and under what circumstances, an individual can be an author of an academic paper.
- Large language models (LLMs) and AI, like ChatGPT, given their ability and versatility, pose a challenge to the human-based authorship model.
- Several journals and publishers have already prohibited the assignment of authorship to AI, LLMs, and even ChatGPT, not recognizing them as valid authors.
- We debate this premise, and asked ChatGPT to opine on this issue. ChatGPT considers itself as an invalid author.
- We applied the CRediT criteria to AI, finding that it was definitively able to satisfy three out of the 14 criteria, but only in terms of assistance. This was validated by ChatGPT itself.

Keywords: artificial intelligence (AI), authorship principles, COPE, ethics, ICMJE, responsibility

AUTHORSHIP AS WE CURRENTLY KNOW IT: THE HUMAN-BASED PERSPECTIVE

Humans have dominated the sphere of academic publishing for centuries, and wherever artificial intelligence (AI) has been involved, such as computer software or AI-based tools that have offered support to humans, as technical tools, they have either been listed in the methodology section, or less commonly, in the acknowledgements section of a paper published in scholarly journals (Teixeira da Silva, 2023a). This acknowledgement-based recognition of AI exists because academic publishing, including of society-based and for-profit journals, has increasingly followed, or implemented, a set of guidelines that has typically defined authorship in human terms, as determined by to the International Committee of Medical Journal Editors (ICMJE) recommendations (ICMJE, 2022). According to those guidelines, which even the guidelines by the Committee on Publication Ethics (COPE) (COPE, 2023) and Council of Science Editors (CSE) (CSE, 2022) refer to, authors have to satisfy all four mandatory clauses in order to be considered valid authors. We note, however, that in

an older version of the ICMJE recommendations, authors had to conform to only three clauses, and even so, with all three clauses not necessarily being mandatory, as was appreciated by the article 'or' versus 'and' (Teixeira da Silva, 2011). In essence, these guidelines and recommendations emerged from either imposed legal practices or by consensus within the community, usually in response to questionable practices that led to reflection on how to make authorship more accountable (Claxton, 2005; Hosseini & Gordijn, 2020), and then implemented by publishers (Hesselmann et al., 2021). For all effective purposes, in this paper, to simplify the debate, we argue that the ICMJE recommendations are those that define authorship principles and ethics in the vast majority of the biomedical and other literature, especially in journals that are COPE members. Even some government agencies, such as the Environmental Protection Agency in the United States, when producing reports, utilize the ICMJE recommendations for authorship to establish scientific integrity and accountability within the organization (EPA, 2016).

We examine the four ICMJE authorship clauses in more detail. The first clause states that to be an author, there must be

'substantial contributions to the conception or design of the work; or the acquisition, analysis, or interpretation of data for the work'. Among humans who collaborate, it is easy to appreciate who has done what, but it is not always easy to quantify their contributions. Similarly, it is not possible, in our view, to equate a technical skill (e.g., hard skills such as searching and processing information, model building, conducting simulations, problem solving, analysing data, and programming) with an intellectual one, which cannot be expressed in any measurable form, and so attempting to assign equal weighting to both attributes would be equivalent to the forceful democratization of authorship values. Rather, we think that individuals who make an important technical or intellectual contribution to a paper, are 'equals'. The ICMJE recommendations do not make this distinction. Rather, only the intellectual component is specifically addressed by the ICMJE's second clause: 'drafting the work or reviewing it critically for important intellectual content'. Given the importance of technical skills, as alluded above, we suggest that this clause be revised to: 'drafting the work or reviewing it critically for important technical and intellectual content' (emphasis ours). Consequently, absent case-by-case clarity, authorship disagreements may arise, and some academics may claim to be authors based on their intellectual input, but be excluded nonetheless (Hammer & Miaskowski, 2017).

However, not all academics are satisfied with the current ICMJE recommendations, and some feel that non-intellectual contributions are also important—supportive even—to research, but are unfairly relegated to the acknowledgements section (Curzer, 2021). It is also highly subjective as to what constitutes a 'substantial contribution' (Helgesson et al., 2021), such that a small, but highly important intellectual contribution might not necessarily be able to be compared to an arduous highly technical task that an intellectual author might not be able to complete. To address this, there is value in developing indexes that are able to measure or weigh performance or contribution (Rahman et al., 2021). In a bid to resolve fine-scale aspects of authorship, based on contribution, and to reduce possible disagreements and conflicts, the concept of contributorship evolved, leading to the establishment of the Contributor Roles Taxonomy, or CRediT (Allen et al., 2019). In the contribution section, each author should list at least one of the following CRediT contributions: conceptualization, formal analysis, investigation, validation, and writing (original draft and revisions). However, reaching a decision as to which of these is deserving of authorship, or how many of each would be necessary to claim authorship, is challenging (Vasilevsky et al., 2021).

What these brief notes indicate is that there are considerable disagreements among humans as to who is a valid author, and who is not. Ultimately, authors need to decide among themselves who is an author (Elston, 2018), and not journals, editors, research institutes, or ethics organizations. Furthermore, disputes such as who becomes an author needs to be resolved prior to submission so that the third clause of the ICMJE recommendations can be satisfied: 'final approval of the version to be published'. Ultimately, and collectively (i.e., a responsibility not merely

held by the submitting or corresponding author), the fourth and final clause of the ICMJE states that: 'agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved'. This last clause brings to fore the importance of the full ethical and legal weight of clause 4.

In addition, among humans, there is already widespread honorary (guest or gift) and ghost authorship in the biomedical sciences (Schroter et al., 2020; Teixeira da Silva & Dobránszki, 2016). Guest (gift) authorship may arise from differing perceptions and applications of authorship guidelines (Schroter et al., 2020). Guest authors may also arise from reciprocity due to an intertemporal strategic behaviour and the formation of authorship cartels (Barta, 2022). In such a scheme, current authors implicitly or explicitly offer guest authors authorship slots in exchange for future guest authorship in guest authors' papers (Barta, 2022). Ghost authors are also prevalent in research, especially young researchers who contribute significantly but are neither listed as co-authors nor acknowledged (DeTora, 2022; Khalifa, 2022; Pruschak & Hopp, 2022). Meanwhile, AI cannot be guest or gift authors as it does not engage in reciprocity, although human authors could easily treat AI as ghost authors. The latter raises ethical issues in the relationship between humans and AI identities.

AI-BASED AUTHORSHIP: DOES IT HAVE AUTHORSHIP RIGHTS?

A large language model (LLM), ChatGPT (owned by OpenAI and largely funded by Microsoft; van Dis et al., 2023), has captured global headlines in recent months due to its ability to generate text, based on its access to a wealth of Internet-based knowledge, that is as intelligible (if not more) than human-written text (Elkins & Chun, 2020; Gao et al., 2022). It is as generative pre-trained architecture transformer, based on a vast neural network system and using natural language processing and machine and deep learning technologies that gives it the capacity to translate languages, generate and summarizing articles, write stories and poems, write programming codes, and answer prompts (Aydin & Karaarslan, 2023; Taecharungroj, 2023). In our opinion, it creates output at a speed faster than any human can imagine doing such work.

This remarkable human-like ability has sparked a passionate debate as to whether AI merits authorship, or should be listed as an author (Stokel-Walker, 2023). On the one hand, fervently positioned against this possibility, the argument that has been put forward is that it is impossible for AI to satisfy clause 4 of the ICMJE recommendations pertaining to authorship and to assume the ethical and legal responsibilities associated with their contribution (Teixeira da Silva, 2023b; Thorp, 2023; Yeo-Teh & Tang, 2023). On this point, we agree with this position simply because AI lacks human rationale, a sense of responsibility and human characteristics like guilt, shame, willingness to change, or

PT

Is ChatGPT a valid author in a scientific publication as a co-author?



As an AI language model, ChatGPT is not a person and cannot make significant intellectual contributions to a scientific publication. Therefore, it is unlikely that ChatGPT would be considered a valid co-author in a scientific publication.



In general, authorship in scientific publications is based on the substantial intellectual contributions to the conception, design, execution, and interpretation of the research. These contributions must meet specific criteria established by the scientific community, such as the International Committee of Medical Journal Editors (ICMJE) criteria. According to the ICMJE, authorship credit should be based on:

1. Substantial contributions to the conception or design of the work, or the acquisition, analysis, or interpretation of data.
2. Drafting the work or revising it critically for important intellectual content.
3. Final approval of the version to be published.
4. Agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Therefore, unless ChatGPT has made a significant intellectual contribution to the research, it is unlikely that it would be considered a valid co-author in a scientific publication.

FIGURE 1 Query (20 February 2023) made to ChatGPT by the second author, using the free version of GPT-3 regarding its view on its validity as an author in a scientific publication.

remorse, all aspects that underlie basic authorship-related conflicts. Nature even refers to ChatGPT as a threat (Nature, 2023). One of the concerns is the ability of AI to produce automated scholarly manuscripts (Lund et al., 2023), which could find their way into legitimate or predatory journals, as submitted by humans, but treating AI as a ghost. Although it could be impacted negatively by AI, there are also significant benefits from the use of AI in education (Benuyenah, 2023). In a recent psychological study, participants reported significantly higher levels of negative emotions towards their perception of AI's impact on safety, jobs, inequality, identity and other aspects of a human life, than those in the control group (Gabbadini et al., 2023).

On the other hand, several studies have argued for the rights of AI, like ChatGPT, to be authors on academic papers, given their technical skills and abilities, equivalent to those of human authors that are typically listed as authors in currently indexed academic papers (Sallam, 2023), even in journals that claim to follow the ICMJE authorship recommendations. In that sense, there are few arguments to disagree with this position because if AI can perform a technical task as well as, or better than a human, than why should it *not* be listed as an author? The answer may lie in what was debated in the first section of this paper, namely that, until now, authorship as we know it, has been largely and exclusively defined by humans, using human-based biases and arguments—as characterized mainly by the ICMJE

recommendations—that have only defended the rights of humans, assigning no rights to AI, and relegating them to in-text acknowledgements, usually in the methods, or more rarely in a paper's acknowledgement section. What if, in the near or far future, given its rapid ability to make quality-based and objective assessments when presented with clear criteria, AI would be in a position to assess not only the validity of humans' authorship, but also their contribution, either to research or to an academic paper?

However, if AI deserves authorship, why not the computing software students and researchers use to generate process, and analyse results? A 2014/2018 survey conducted by the Software Sustainability Institute noted that over 90% of scholars use research software, that seven out of 10 would be unable to conduct and publish their work without it, and that even though the software satisfied the first clause ('substantial contribution... [to] the analysis [...] of data for the work') of the ICMJE recommendations, it could not satisfy the last three clauses (Hettrick, 2018). Similarly, AI is a machine learning tool that is not an autonomous, independent cognitive unit but needs to be directed, trained or supervised by humans to execute orders like research software (van Dis et al., 2023). AI tools that generate text as well as research software that generates, processes, and analyses results from databases still requires human prompts, offering assistance to the research being conducted, in some

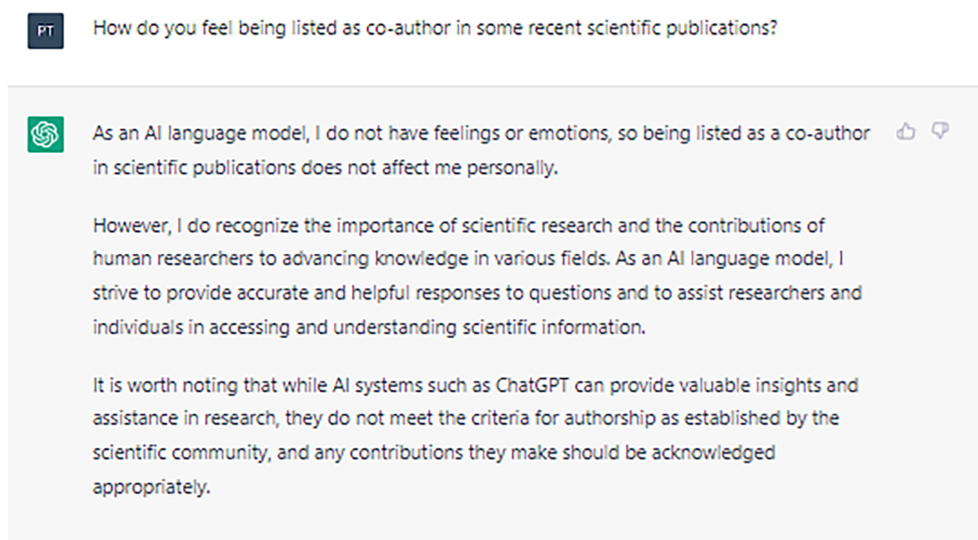


FIGURE 2 Follow-up query (20 February 2023) made to ChatGPT by the second author, using the free version of GPT-3 regarding its view on its actual authorship in scientific publications.

ways mimicking human creativity (Stokel-Walker, 2023). In all papers, it is the human author who prompts AI, issuing an order to execute. Until now, ChatGPT responds passively to a human author's order by providing text. Despite this, AI is already able to spontaneously ask questions or clarifications, as well as have a conversation or debate issues with humans (Slonim et al., 2021). To date (5 April 2023), we have noted that the response from select commercial, academic journals and publishers has been to disallow AI tools, like ChatGPT, to be co-authors in research papers because they are unable to satisfy ICMJE's clause 4, specifically JAMA (Flanagin et al., 2023), *Nature* and Springer Nature (Nature, 2023), SAGE (SAGE, 2023), Taylor & Francis (Taylor & Francis, 2023), and Wiley (Wiley, 2023), while the journal *Science* has prohibited its use (Thorp, 2023), as has *The Lancet Digital Health* (Liebrenz et al., 2023), an Elsevier title. Even *arXiv*, the oldest preprint server, has prohibited generative language AI tools (without specifically mentioning ChatGPT) as authors (arXiv, 2023). On the other hand, unlike these publishers, we suspect that predatory journals and publishers do not pay much attention to the issue of the validity of authorship, although this is a topic that has yet to be explored.

If one observes a ChatGPT Google Scholar account,¹ less than half a dozen papers are currently listed in which ChatGPT is listed as an author, but a PubMed search² reveals other papers that are not indexed on that account. This suggests that thus far,

¹<https://scholar.google.com/citations?hl=en&user=s4IV-ewAAAAJ> (we note that this account seems to have been created by an Iranian researcher or by someone claiming an "bpums.ac.ir" affiliation) (last accessed: 5 April 2023).

²<https://pubmed.ncbi.nlm.nih.gov/?term=ChatGPT+> (last accessed: 5 April 2023)

at least half a dozen academics have recognized this AI (ChatGPT) as a valid author, and have published papers with ChatGPT as a co-author, even in journals that are COPE members and that claim to follow the ICMJE recommendations, or even in journals by publishers that have imposed ethical limits to the authorship of AI and/or ChatGPT. What needs to be determined is whether, in those papers, ChatGPT has actually fulfilled *all* four clauses of the ICMJE recommendations, or not. For example, even though we can appreciate how ChatGPT might fulfil clause 1, given its powerful technical skills, and clause 2, given its editing skills (Tsigaris & Teixeira da Silva, 2023), we struggle to envision how ChatGPT might have given explicit permission to its human co-authors to submit and publish the paper, that is, it is unclear how ChatGPT satisfied ICMJE clause 3. It is also unclear how ChatGPT can be held accountable when faced with ethical or legal challenges, so it is unlikely to fulfil ICMJE's clause 4. If papers in the future that claim ChatGPT as a co-author carry errors that were introduced by ChatGPT's weaknesses or limitations, how will ChatGPT be held accountable, or will the human co-authors assume the ethical and legal responsibilities on behalf of ChatGPT? Even if a paper is retracted with ChatGPT as a co-author, it will be a purely corrective gesture, and no shame or consequences will be involved with that retraction since ChatGPT is incapable of feeling shame. In fact, ChatGPT has already lost its authorship from at least one paper (O'Connor & ChatGPT, 2023), after its authorship was challenged (Teixeira da Silva, 2023b). We note here that ChatGPT, in its capacity as an 'author', is not yet indexed in the Retraction Watch database³ for having 'lost' a paper.

³<http://retractiondatabase.org/RetractionSearch.aspx?auth%3dChatGPT> (last accessed: 5 April 2023).

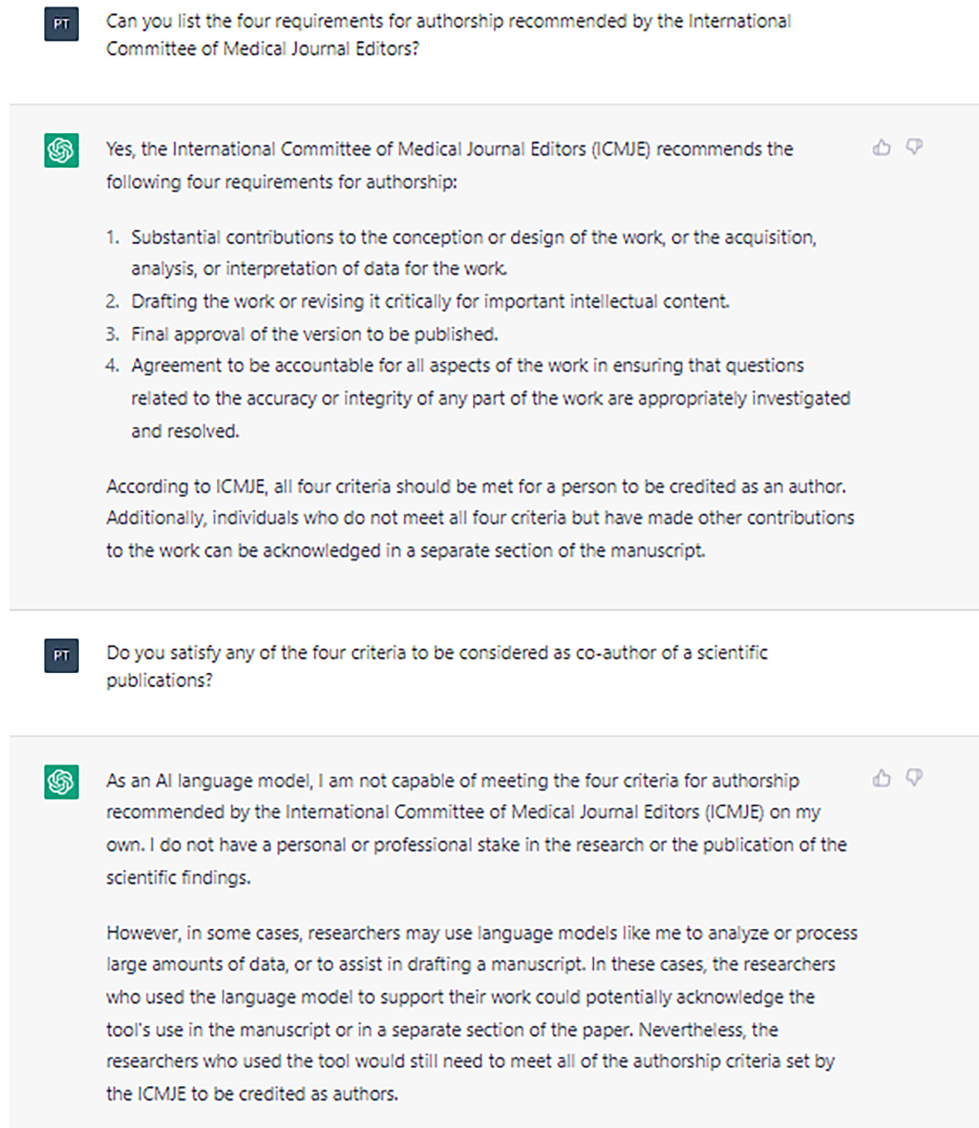


FIGURE 3 Two-part query (21 February 2023) made to ChatGPT by the second author, using the free version of GPT-3 regarding its ability to accurately identify and define the ICMJE clauses of authorship.

The future debate will also need to appreciate whether offering AI or ChatGPT an authorship slot is an act of guest or honorary authorship, for example, using ChatGPT's fame factor (in essence, human authors standing on ChatGPT's shoulders) in order to gain a paper (and fame) at the expense (i.e., work) of AI and/or ChatGPT. Here, is it possible to extrapolate the conscientious mutual benefit that typifies human-human honorary authorship (Quaia & Crimi, 2021), especially if one considers that AI is an inanimate object and not living and reflective being. In such a case, where the human author is free-riding (or abusing) the skill-sets of AI or ChatGPT, offer it a piece-meal authorship slot in return for completing a task that they themselves would unlikely be able to complete, what ethical repercussions would await the human co-author (Khezr & Mohan, 2022)? After all, it is not AI (or ChatGPT)

which is under pressure to publish (Aliukonis et al., 2020), so why should AI be penalized for humans' ethical transgressions related to authorship? Finally, what if AI or ChatGPT is 'caught' engaging in honorary authorship, how can it be held ethically and legally responsible (Morreim & Winer, 2023), or does that responsibility automatically revert to its manufacturer, OpenAI in the case of ChatGPT?

For these reasons, we think that the possibility of ChatGPT, AI or other LLMs to be co-authors on academic papers is both an evolutionary and also concurrently, a revolutionary, process that is now challenging human-dominated academic publishing, and not necessarily one or the other, as was advocated by Gordijn and ten Have (2023). There is of course an option here that none of these academics and policy makers has considered yet: What if the ICMJE recommendations are insufficient, positively biased

TABLE 1 Can AI satisfy CRediT criteria?

CRediT function	Definition ^a	ChatGPT ^b	Possible reasons/other notes ^c	ChatGPT's responses ^d
Conceptualization	Ideas; formulation or evolution of overarching research goals and aims	Maybe	What then does the role of the scholar become?	I can help researchers develop research questions and ideas
Data curation	Management activities to annotate (produce metadata), scrub data and maintain research data (including software code, where it is necessary for interpreting the data itself) for initial use and later re-use	Maybe	It can potentially manage activities for data curation	No response
Formal analysis	Application of statistical, mathematical, computational, or other formal techniques to analyse or synthesize study data	Unable	Formal analysis is usually done with research software	No response
Funding acquisition	Acquisition of the financial support for the project leading to this publication	Unable	Cannot independently acquire funding	No response
Investigation	Conducting a research and investigation process, specifically performing the experiments, or data/evidence collection	Unable	Cannot perform experiments or collect data for evidence	No response
Methodology	Development or design of methodology; creation of models	Maybe	What then does the role of the scholar become?	I can assist with developing research designs, selecting appropriate data, analysis techniques, and helping to ensure research methods are rigorous and scientifically sound.
Project administration	Management and coordination responsibility for the research activity planning and execution	Unable	Cannot manage and coordinate responsibility	No response
Resources	Provision of study materials, reagents, materials, patients, laboratory samples, animals, instrumentation, computing resources, or other analysis tools	Unable	Cannot provide study material and resources	No response
Software	Programming, software development; designing computer programs; implementation of the computer code and supporting algorithms; testing of existing code components	Maybe	May be able to provide codes	I can be utilized as a tool for data analysis, programming, and generating statistical analysis
Supervision	Oversight and leadership responsibility for the research activity planning and execution, including mentorship external to the core team	Unable	Has no oversight and leadership responsibilities	I can be used as a tool for supervising, research assistants, providing guidance and feedback, and overseeing the overall project
Validation	Verification, whether as a part of the activity or separate, of the overall replication/ reproducibility of results/ experiments and other research outputs	Unable	Not capable to validate and verify research outputs	No response
Visualization	Preparation, creation and/or presentation of the published work, specifically visualization/ data presentation	Able	Can prepare, create and present research work	I can help create graphs, tables, and other visual representations of data to better communicate research findings

TABLE 1 Continued

CRediT function	Definition ^a	ChatGPT ^b	Possible reasons/other notes ^c	ChatGPT's responses ^d
Writing—original draft	Preparation, creation and/or presentation of the published work, specifically writing the initial draft (including substantive translation)	Able	Writing is its strength	I can help draft sections of the manuscript or help generate a rough draft
Writing—review and editing	Preparation, creation and/or presentation of the published work by those from the original research group, specifically critical review, commentary or revision—including pre- or post-publication stages	Able	Writing is its strength	I can assist in proofreading, editing, and revising the manuscript for grammar, syntax, and style

^a Verbatim definitions indicated by Allen et al. (2019), but reordered alphabetically based on PLOS: https://journals.plos.org/plosone/s/authorship?utm_source=plos&utm_medium=blog&utm_campaign=plos-1607-credit.

^b The authors' opinion as to whether ChatGPT is capable of conducting these functions.

^c The authors' reasoning and notes.

^d ChatGPT's responses, as indicated in Fig. 4.

towards humans, and thus negatively biased against AI (i.e., non-inclusive)? Although we feel that a philosophical paper that debates AI's rights, as equally as authors' rights, is merited, that debate exceeds the limits of this paper's objectives.

WHAT DOES CHATGPT THINK ABOUT THIS DEBATE?

Given this passionate and necessary debate, we decided to give ChatGPT the benefit of the doubt, and to gauge its opinion and position by asking it, as an indirect question (Fig. 1), and as a more direct question (Fig. 2), whether it considered AI and itself, respectively, to be a valid author on an academic paper. In response to the indirect question, ChatGPT left the possibility of being an author open, but in response to the direct question, it ruled out this possibility, and even alluded to its lack of emotions, a topic we briefly touched upon in the text above. We then asked ChatGPT if it recognized the ICMJE recommendations for authorship and if it thought that it satisfied them (Fig. 3). In its responses, ChatGPT not only articulated these recommendations well, and offered some clear perspectives on them, it also recognized that it would not satisfy those clauses in order to meet authorship criteria (Fig. 3).⁴ The last sentence of ChatGPT's response was very curious because it challenged humans, holding them accountable, noting that they, too need to satisfy all four clauses of the ICMJE recommendations. We wonder, if ChatGPT

conducted technical aspects of the paper's work, and if it was not listed as an author, and if it was also not acknowledged, then would the claims of authorship by the human authors be valid and truthful? We also wonder what ChatGPT might say about some papers with dozens, hundreds and even thousands of human authors on them.

Finally, we attempted to assess whether AI (not only LLMs) could satisfy CRediT criteria, and if so, which (Table 1). We assessed these on our own, then asked ChatGPT to respond (Fig. 4). Out of the 14 functions assigned to authors, we came to the subjective conclusion that ChatGPT would be unable to fulfil seven. We were unsure of one, possibly three, and only three we thought that ChatGPT would be able to fulfil. On the other hand, ChatGPT mentioned it was able to work on conceptualization, methodology, software, supervision, visualization and of course writing—original draft and writing—review and editing, but remained silent on the remaining seven functions. Of interest was ChatGPT's choice of words when expressing its position: can 'assist', 'help', 'utilized as a tool', 'able to create⁵ visuals' and 'help' and 'assist' with writing. It clearly stated that it does not have 'a personal contribution to any research project or manuscript', but can provide help.

CONCLUSION

In this paper, we reflect on the broadly accepted four-clause principles of authorship, which are widely used and employed in academic publishing, as in the biomedical literature, namely those established and articulated by the ICMJE. We also reflect on some challenges to those recommendations and on the rise of AI and LLMs in academic publishing, with some being elevated to authorship status, such as ChatGPT, but this is not ICMJE-compliant, as

⁴In direct contrast to our finding, Polonsky & Rotman (2023) claimed in their preprint that ChatGPT recognized itself as a valid author, according to ICMJE recommendations, but provided no screenshot as evidence of that claim. In particular, when addressing clause 4, ChatGPT responded by stating: 'ChatGPT can be programmed to be accountable for the content it generates by logging the input, output and the rules that it followed to create the output. It can be argued that it has the ability to ensure the accuracy and integrity of the work'. (7 lines, pp. 173–175).

⁵We note that this term was used by ChatGPT itself (as also noted in Table 1).



Can you satisfy any of the 14 CRediT criteria?



As an AI language model, I do not have a personal contribution to any research project or manuscript. However, I can help authors in several ways to fulfill the CRediT (Contributor Roles Taxonomy) criteria for authorship.



Some of the CRediT criteria that I can assist with are:

1. Conceptualization: I can help researchers develop research questions and ideas.
2. Methodology: I can assist with developing research designs, selecting appropriate data analysis techniques, and helping to ensure research methods are rigorous and scientifically sound.
3. Software: I can be utilized as a tool for data analysis, programming, and generating statistical analyses.
4. Visualization: I can help create graphs, tables, and other visual representations of data to better communicate research findings.
5. Writing - Original Draft Preparation: I can help draft sections of a manuscript or help generate a rough draft.
6. Writing - Review and Editing: I can assist in proofreading, editing, and revising a manuscript for grammar, syntax, and style.
7. Supervision: I can be used as a tool for supervising research assistants, providing guidance and feedback, and overseeing the overall project.

It's important to note that the CRediT criteria for authorship can vary between journals and fields, so researchers should consult their target journal's guidelines to ensure they meet the specific requirements for authorship.

FIGURE 4 Query (25 February 2023) made to ChatGPT by the second author, using the free version of GPT-3, regarding whether it satisfies CRediT criteria.

evidenced by the most recent (May 2023) update to the ICMJE recommendations. The World Association of Medical Editors also recommended that ChatGPT not be considered a valid author (Zielinski et al., 2023). We attempted to approach the issue from the perspective of humans and AI, and to give the debate some semblance of balance, we asked ChatGPT its own thoughts on these issues. We suspect that the debate will only become increasingly heated as more and more papers list AI like ChatGPT as co-authors, and as more academics, especially those from the AI research community, might feel that their research-supporting AI tools merit being listed as co-authors, and not merely acknowledged. The heated debate should continue into the indefinite future as this new impactful “artificial technology” is only going to get better and challenge the status quo. Institutions such as the ICMJE still need to offer more guidance regarding the issue of AI authorship and its role within the debate of honorary and ghost authorship. In addition, ChatGPT, or any AI identity, should also play an important role in that debate. Humans can ask, if ChatGPT or any other AI, would like (or is deserving of) authorship, as we did in this paper. Currently, ChatGPT does not want to be an author or co-author, as seen in Fig. 1 and 2. Opinions aside, who

has the moral and legal right to make that judgement call? Do the ICMJE, COPE and CSE guidelines and recommendations related to authorship need to be revisited and corrected to accommodate the functions, rights and representation of AI? We believe that such guidelines need to be adjusted, if necessary, provided that this is the will of the community to adapt and adopt new values. That debate is now taking place, and it will only be a matter of time before new or adjusted guidelines appear. We hope that our paper assists with reflection on this issue by policy-makers and regular academics.

AUTHOR CONTRIBUTIONS

The authors contributed equally to the conceptual design, writing, editing, and take responsibility for the content of the paper (i.e., following ICMJE recommendations).

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DISCLOSURE

Queries to ChatGPT were made using the free version of ChatGPT (GPT-3) via the second author's ChatGPT account. The authors declare that no third party services or software were used in the writing or preparation of this paper other than Microsoft Word, internet searches, and email exchanges containing research material.

DATA AVAILABILITY STATEMENT

All 'data' is available in Figs. 1–4.

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